pollution from the legend or Prandtl number annotation. With those bounds set, they can be safely ignored. Once the undesirable annotations are ignored, the three profiles can begin to be separated. Each non-white pixel is assigned to one of three brightness classes. Once each non-white pixel is assigned to one of the three groups, a median filter is applied. This filter looks at each column of pixels, and assigns a representative pixel coordinate for each of the three brightness groups by taking the median verical location of pixels in each group for that collumn. For places where curves cross or overlap, Gaussian smoothing is applied to improve the resultant profile.

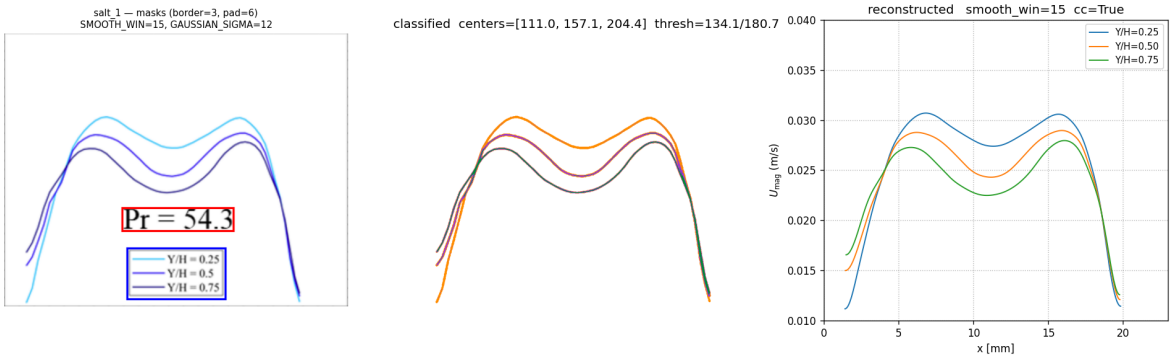


Figure 4.3: Masking and Extraction Images for Salt Test 1

To verify that the digitization is serviceable, the saved array of data is then overlaid back upon the source image using the calibrated coordinate frame to provide visual confirmation.

Error Quantification Against Salt Data

Since the salt tests have a raw full-field velocity dataset, the accuracy of the entire digitization process can be bounded for factoring in to the uncertainty analysis of the water full-field velocity data comparisons. For each salt test and height, the raw data was sliced at the corresponding Y/H row and the digitized curve was restricted to the overlapping horizontal window. The comparison involves roughly 80 data points across the board for each height comparison. L_2, bulk excludes an additional 1.25 mm from either side of the overlap window to provide a frame of reference for the accuracy of the digitization away from either tail of